

Units Conversions & Measurements

Problem Sheet

Warm-Up Problems

Problem 1

Convert the following values into the requested units. Show your work.

- a) 5 miles into kilometers. (Use 1 mile = 1.609 km)
- b) 3 feet into inches. (Use 1 foot = 12 inches)
- c) 2.5 hours into seconds.

Problem 2

The speed limit on a road is **60 miles per hour**.

- a) Convert this speed into meters per second. (Use 1 mile = 1609 m, 1 hour = 3600 s)
- b) Convert the same speed into kilometers per hour.

Problem 3

A water bottle has a volume of **2 liters**.

- a) Express this volume in milliliters.
- b) Express the same volume in cubic centimeters.
- c) Express the same volume in U.S. fluid ounces. (Use 1 L $\approx$ 33.8 fl oz)

Real Problems

Problem 1

“— Give him five knuts,” said Hagrid sleepily.

“— Knuts?”

“— Little bronze ones.”

Harry counted out five little bronze coins, and the owl held out his leg so Harry could put the money into a small leather pouch tied to it. Then he flew off through the window.

— J. K. Rowling

The approximate value of a **galleon** is **5 British pounds sterling**. On the day Harry went to the bank, **1 British pound = 1.26 U.S. dollars**. Currency relations in the wizarding world: **1 galleon = 17 sickles**, and **1 sickle = 29 knuts**.

- a) **How much money in U.S. dollars did Harry give to the owl?** Round your answer to the nearest cent. Give your answer as two numbers: the number of whole dollars and the number of cents.
- b) **How much does Harry’s wand cost in U.S. dollars if he bought it for 7 galleons?** Round your answer to the nearest cent. Give your answer as two numbers: the number of whole dollars and the number of cents.

Problem 2

From the starting line, **Mickey** ran toward the finish at a speed of **6.7 miles per hour**, while **Goofy**, who started at the same time as Mickey, ran toward the finish at a speed of **8.2 feet per second**.

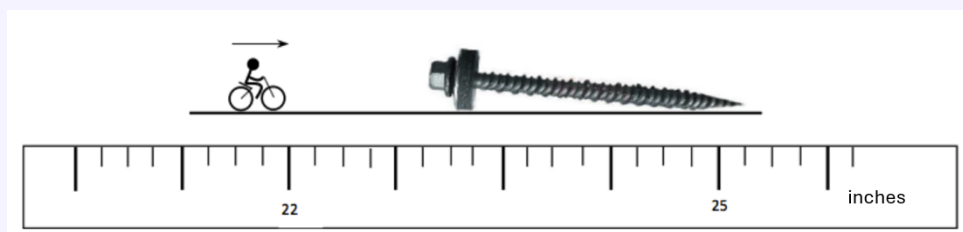
Who reached the finish line first?

Problem 3

Using the figure, determine how much time **Tom** on a mini-bicycle takes to pass by the screw lying on the table, if **1 inch = 2.54 cm**. Tom rides at a constant speed of **0.6 miles per hour**. Give your answer in seconds, rounded to the nearest hundredth.

Can Tom on the bicycle be considered a *point mass* in this situation?

Note. A point mass is a body whose dimensions can be neglected under the conditions of the given problem.



Problem 4

While vacationing on a tropical island, **Bob** rented a scooter. The main scale of its speedometer was calibrated in the locals' traditional unit of speed — “**alligators per minute**.”

The rental owner, wishing to accommodate European tourists, learned that in the metric system, speed should be measured in **meters per second (m/s)**. So, next to the local scale, he added a second “standard” European scale (see figure).

Determine:

- What is the maximum speed of the scooter's exotic scale, expressed in kilometers per hour?
- What is the speedometer reading on the figure, expressed in kilometers per hour?
- What is the length of the local “alligator,” expressed in meters?

